RailSmart - UK Railway Intelligence Platform

3-Month Project Timeline

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Timeline Overview

- Month 1 (Weeks 1-4): Data Foundation, Cleaning & Initial Analysis
- Month 2 (Weeks 5-8): Deep Analysis & Feature Engineering
- Month 3 (Weeks 9-12): Forecasting, Dashboard & Final Delivery

Detailed Timeline

Phase Name	Week	Focus of the Phase	Deliverables
Phase 1: Data Model & Data Understanding	Week 1	Project scope definition, data source identification, initial data exploration, data dictionary creation	Data model design, Project plan
Phase 2: Data Cleaning & Preprocessing	Week 2-3	Clean and preprocess data, handle missing values, outlier detection, data type conversions, data validation	Cleaned dataset ready for analysis, Data preprocessing notebook
Phase 3: Exploratory Data Analysis (EDA)	Week 4-5	Statistical analysis, pattern identification, correlation analysis, visualization of key metrics	EDA insights and visualizations
Phase 4: Analysis Questions Phase	Week 6-7	Determine all possible analysis questions from the dataset, perform deep analysis, answer business questions	Set of analysis questions with answers, Analysis documentation
Phase 5: Feature Engineering	Week 8	Create derived features, route analysis, delay pattern analysis, price trend analysis, prepare data for ML	Feature engineering notebook
Phase 6: Forecasting Questions Phase	Week 9-10	Predict number of rides for next month, forecast revenue for each day, predict demand on different ticket classes, build and validate ML models	Forecasting models, Visualization plots answering forecasting questions
Phase 7: Visualization Dashboard Development	Week 11	Build Tableau visualization dashboard, create interactive charts, design user-friendly interface	Tableau dashboard visualizing all analysis and forecasting results
Phase 8: Final Presentation & Documentation	Week 12	Prepare final report and presentation, document all findings, create comprehensive project summary, code documentation	Final report, Project presentation, Complete documentation

Tools & Technologies by Phase

Weeks 1-3: SQL, Python (pandas, Matplotlib)

Weeks 4-7: SQL, Python (pandas, Matplotlib, Seaborn)

Weeks 8-10: Python (scikit-learn, pandas, Matplotlib, Prophet)

Weeks 11-12: Tableau, SQL, Python (pandas, Matplotlib)

Key Deliverables

1. Cleaned Dataset - Preprocessed and ready for analysis

2. Data Preprocessing Notebook - Jupyter notebook with all cleaning steps

3. Analysis Questions & Answers - Comprehensive dataset analysis

4. Forecasting Models - Ride predictions, revenue forecasts, demand predictions

5. Tableau Dashboard - Interactive visualization dashboard

6. Final Report & Presentation - Complete project documentation